

ENGINEERING · SHIELDING SYSTEMS · QUANTUM SENSING

SHIELDED CHAMBER BLUEPRINT

Phase 0 Configuration: Design Specifications for
Consciousness-Dependent Quantum Decoherence Measurements

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The Silence Paradigm Project

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SECTION I

Design Objective

This document specifies the engineering requirements for a shielded experimental chamber designed to test whether the cognitive state of a conscious observer produces measurable changes in the decoherence rate (T2) of a nearby quantum system. The chamber creates a controlled environment of maximum electromagnetic, acoustic, vibrational, and thermal isolation while maintaining a symmetric geometry optimized for minimal boundary-induced decoherence.

The independent variable is the observer's cognitive state (seven controlled conditions C1 to C7, plus an empty-chamber instrumental baseline, EEG-verified). The dependent variable is the coherence time T2 of the quantum sensor. The chamber must reduce all known environmental coupling channels below the threshold where standard decoherence models predict T2 limits, such that any residual variation across consciousness conditions cannot be attributed to uncontrolled environmental factors.

1.1 Performance Requirements

Parameter	Specification
Electromagnetic attenuation	> 120 dB across 10 Hz to 10 GHz
Magnetic field attenuation	> 80 dB DC and AC (50/60 Hz)
Acoustic isolation	> 80 dB; interior ambient < 20 dB SPL
Vibration isolation	< 10^{-6} m/s ² across 1 to 100 Hz at sensor platform
Thermal stability	+/- 0.01 K over 60-minute session (with active control)
Interior geometry	Perfect cube, 3.0 m x 3.0 m x 3.0 m clear volume
Observer capacity	Single seated adult at geometric center

Table 1.1: Chamber performance requirements.

Magnetic target. A requirement of > 60 dB magnetic attenuation is under-specified for the NV sensor's sensitivity to mains-frequency (50/60 Hz) field fluctuations. Typical lab AC magnetic noise is ~100 nT; 60 dB attenuation leaves ~100 pT residual, which approaches NV T2 limits at ensemble scale. The > 80 dB target requires a dual-layer mu-metal architecture with an air gap (see Section 4 and Section 5.2).

SECTION II

Overall Architecture

2.1 Dimensions

Parameter	Specification
Inner clear volume	3.0 m x 3.0 m x 3.0 m (27.0 m ³)
Outer footprint	4.3 m x 5.5 m x 4.3 m (incl. airlock; +100 mm for dual mu-metal)
Total shielding thickness	650 mm per side (dual-layer magnetic adds 50 mm)
Orientation	Entry on east face; observer faces west when seated
Primary access	East face, dual-door airlock entry system (see Section 3)
Outer door	1.0 m W x 2.1 m H, east face of airlock
Inner door	1.0 m W x 2.1 m H, west face of airlock (enters main chamber)
Door seals (both)	Multi-layer: RF gasket + dual mu-metal overlap + acoustic compression seal
Feedthrough panel	West face (opposite entry), centered
Total chamber mass (est.)	10,500 to 14,500 kg (incl. airlock and second mu-metal layer)
Foundation requirement	Reinforced concrete slab, 300 mm min, vibration-isolated

Table 2.1: Overall chamber dimensions.

2.2 Geometry Rationale

The perfect cubic geometry is selected for four engineering reasons:

- Spatial symmetry: Minimizes anisotropic boundary conditions that could produce position-dependent decoherence variation.
- Mode density: Cubic geometry supports the lowest and most uniform electromagnetic mode density for a given volume.
- Fabrication simplicity: Flat panels simplify shielding layer construction, seam welding, and quality assurance.
- East-facing entry: Positions the observer facing west (toward the feedthrough panel / deepest interior), maximizing the distance between the access opening (highest seal-leakage risk) and the primary sensor array.

2.3 Coordinate System

Origin at inner floor corner (southeast). X-axis west, Y-axis north, Z-axis vertical. Geometric center at (1.5, 1.5, 1.5) m. The observer enters from the east (X = 0) and is seated at center facing west (X = 3.0). The feedthrough panel is on the west face (X = 3.0). All sensor and observer positions referenced to this coordinate system.

SECTION III

Airlock Entry System

A dual-door airlock anteroom is positioned on the east face of the main chamber. The airlock serves three critical engineering functions: (1) it prevents direct line-of-sight RF leakage when either door is open, (2) it provides a contamination buffer where the observer's electromagnetic emissions from clothing, devices, and body can be checked and stabilized before entering the shielded interior, and (3) it ensures the observer passes sequentially through all five isolation layers, with no bypass path.

3.1 Airlock Dimensions

Parameter	Specification
Interior volume	1.0 m (E-W) x 1.2 m (N-S) x 2.3 m (height)
Outer door (east face)	1.0 m W x 2.1 m H, opens outward
Inner door (west face)	1.0 m W x 2.1 m H, opens inward to main chamber
Interlock	Only one door can be open at a time (electrical interlock + mechanical latch)
Shielding	All five layers continuous through airlock walls, ceiling, and floor
Lighting	Battery-powered LED (shielded), manual switch inside airlock
Communication	Fiber-optic intercom to control room, active in airlock

Table 3.1: Airlock dimensions.

3.2 Entry Protocol

- Step 1, Pre-entry check: Observer removes all metallic objects, electronic devices, and magnetically contaminated clothing in the preparation area outside the chamber. EEG cap is fitted and verified. Wireless radios on the EEG amp (Bluetooth, Wi-Fi) are confirmed disabled in hardware; unit operates battery-only with fiber-optic data egress.
- Step 2, Outer door open: Observer enters airlock. Outer door is sealed behind the observer.
- Step 3, Stabilization: Observer remains in airlock for 60 seconds. RF power meter inside airlock confirms observer's residual electromagnetic emission is below threshold (< -80 dBm across 10 kHz to 10 GHz), including emission from the EEG amp itself. Fluxgate magnetometer confirms no ferromagnetic contamination.
- Step 4, Inner door open: Once contamination check passes, inner door is released. Observer enters main chamber and proceeds to seated position at geometric center.
- Step 5, Inner door sealed: Inner door is closed and sealed. Both doors now closed. Full shielding integrity restored.

- Step 6, Settling period: Minimum 15-minute ingress settling period (data discarded) for observer postural oscillation damping on the vibration-isolated floor and thermal re-equilibration of the chamber after ingress. This ingress settling is distinct from, and precedes, the 5-minute pre-condition buffer specified in Section 6.1.

3.3 Exit Protocol

Reverse sequence: inner door opens first (observer exits to airlock), inner door sealed, outer door opens, observer exits. At no point are both doors open simultaneously. Emergency override (panic bar) on inner door bypasses interlock for immediate egress.

Panic-bar event, session invalidation. Any activation of the inner-door panic bar mechanically bypasses the airlock interlock and creates a transient direct-RF path between the exterior environment and the sensor volume. This is correct for observer safety. However, any such event invalidates the current session and requires full shielding reverification (see Section 12.1) before the next session. Sessions in which the panic bar was not used but was contacted or tested are logged but not invalidated.

Sequential layer passage. The airlock ensures that the observer's entry path passes through every isolation layer in sequence: Layer 5 (outer Faraday), Layer 4 (vibration), Layer 3 (copper RF), Layer 2b (outer mu-metal), Layer 2a (inner mu-metal), Layer 1 (acoustic/thermal), interior. No layer is bypassed. This is critical for maintaining shielding integrity and preventing contamination bridging.

SECTION IV

Five-Layer Isolation System

The isolation system consists of five functional layers, with the magnetic shielding implemented as a dual-layer subsystem (2a inner, 2b outer) separated by an air gap. Layers are numbered from innermost (Layer 1) to outermost (Layer 5). Total thickness per side: 650 mm.

Layer	Function	Material	Thick.	Attenuation
1 (Inner)	Acoustic + active thermal	Acoustic foam + aerogel + Peltier-cooled liner	150 mm	> 40 dB acoustic, +/- 0.01 K (active)
2a	Magnetic, gradient suppression	Mu-metal (80% Ni alloy), inner shell	3 mm	Combines with 2b for > 80 dB
2b	Magnetic, primary attenuation	Mu-metal (80% Ni alloy), outer shell + 50 mm air gap	3 mm + 50 mm gap	> 80 dB DC/LF magnetic (combined with 2a)
3	RF / microwave shielding	OFHC copper sheet, welded seams	2 mm	> 100 dB (10 kHz to 10 GHz)
4	Vibration isolation	Pneumatic isolators + neoprene dampers	200 mm	$< 10^{-6}$ m/s ² (1 to 100 Hz)
5 (Outer)	Structural + outer Faraday	Aluminum shell + copper mesh cage	245 mm	> 120 dB combined EM

Table 4.1: Five-layer isolation system summary.

Layer ordering rationale. The dual-layer magnetic system (2a/2b) sits inside the copper RF shell (Layer 3) for two reasons: (i) mu-metal permeability is degraded by high RF field exposure, so the copper shell pre-attenuates incident RF before it reaches the mu-metal; (ii) the air gap between 2a and 2b increases effective magnetic shielding by a factor of ~3x compared to a single shell of equivalent total thickness. The acoustic/thermal layer (Layer 1) remains innermost to establish the stable acoustic and thermal environment the observer and sensors experience, and is an actively controlled layer rather than passive.

SECTION V

Layer-by-Layer Material Specifications

5.1 Layer 1: Acoustic and Active Thermal Isolation

Component	Material / Specification	Notes
Acoustic panels	Melamine closed-cell foam (BASF Basotect or equiv.), 100 mm, NRC > 0.95, Class A fire	Acoustic specialty
Thermal blanket	Silica aerogel blanket (Aspen Cryogel or equiv.), 50 mm, $k < 0.015$ W/mK	Passive component
Active liner	Perforated aluminum sheet, 1 mm, 20% open area, with integrated Peltier loop and chilled-fluid manifold	Dissipates observer heat load; bi-directional ± 0.01 K
Chiller	External recirculating chiller, ± 0.05 K output, sealed glycol loop, located in equipment room	Vibration-decoupled via flexible fluid lines through waveguide-below-cutoff penetration
Mounting	Non-magnetic stainless clips (316L SS), bonded to foam with silicone adhesive	Non-ferromagnetic

Table 5.1: Layer 1 acoustic and active thermal isolation.

Active thermal control. A passive Layer 1 (foam + aerogel only) is insufficient. A seated adult dissipates ~ 100 W; EEG electronics, lighting, and the fluorescence laser add further heat. In a sealed aerogel-lined chamber with no air exchange, this produces an interior temperature rise of ~ 1 to 3 K over a 70-minute session, two orders of magnitude worse than the ± 0.01 K requirement. Active control is therefore mandatory. The Peltier-cooled liner removes observer heat at the generation point; the external chiller dumps the heat outside the chamber via fluid loop through a vibration-decoupled waveguide penetration.

5.2 Layer 2: Dual Mu-Metal Magnetic Shielding

Component	Specification	Notes
Primary shields (2a and 2b)	Mu-metal (MuShield or Magnetic Shield Corp.), 3 mm sheet, $\mu_r > 80,000$ initial, $> 300,000$ max	Both shells hydrogen-annealed after forming
Air gap	50 mm radial gap between 2a and 2b, maintained by non-magnetic standoffs	Increases effective shielding by $\sim 3\times$
Seam treatment	Mu-metal overlap strips, 50 mm overlap at all joints, spot-welded	Continuous magnetic path on both shells
Penetrations	Mu-metal waveguide-below-cutoff tubes, ID 25 mm, length 250 mm min	Through both shells, coaxial
Door overlap	Mu-metal finger stock + overlap flange, 100 mm overlap, both shells	Dual-row spring-loaded contact

Table 5.2: Layer 2 dual mu-metal magnetic shielding.

Hydrogen annealing. All mu-metal components (both shells) must be hydrogen-annealed at ~1,150 C after final forming and before installation. Mechanical stress from cutting, bending, or drilling degrades permeability by up to 90%. Annealing restores full magnetic shielding performance. Budget 6 to 12 weeks lead time for annealing service (longer for European vendors).

5.3 Layer 3: RF / Microwave Shielding

Component	Specification	Notes
Shield panels	OFHC copper sheet (C10100), 2 mm, oxygen-free high-conductivity	Conductivity > 100% IACS
Seam joining	Continuous TIG or silver-solder, all seams welded	No mechanical-only joints
Ground bonding	Copper braid to building ground, 25 mm ² min cross-section	Single-point ground
Penetrations	Waveguide-below-cutoff copper tubes, ID 20 mm, length 200 mm	Bonded to copper shell
Door contact	BeCu RF finger stock gasket, continuous around full perimeter	Double row for redundancy

Table 5.3: Layer 3 RF / microwave shielding.

5.4 Layer 4: Vibration Isolation

Component	Specification	Notes
Active isolators	Pneumatic air-spring isolators (TMC or Newport), 4 to 8 units, 2,500 kg cap. ea.	Up rated for added mass
Passive dampers	Neoprene/rubber pads, Shore A 40 to 60, 50 mm	Between frame and floor
Isolation frame	Welded steel frame, 100 mm x 100 mm box section	Supports all inner layers
Sensor sub-platform	Granite or cast-iron optical table, 1.5 m x 1.0 m x 0.3 m, mass > 500 kg	Separate isolators from main
Flexible couplings	Bellows / flex conduits on all connections between inner and outer	Prevent vibration bridging

Table 5.4: Layer 4 vibration isolation.

5.5 Layer 5: Structural Shell and Outer Faraday Cage

Component	Specification	Notes
Structural shell	6061-T6 aluminum plate, 5 mm, welded frame	Mechanical envelope
Faraday mesh	Copper mesh, 100 x 100 per inch, bonded to inner face	Cost & fragility consideration
Structural frame	Aluminum extrusion, 80 mm x 80 mm T-slot profile	Modular assembly
Floor	12 mm aluminum plate on steel frame, level to +/- 1 mm over 4.3 m	Anti-static coating
Access panel	Removable panel opposite door, for equipment installation	Sealed after setup

Table 5.5: Layer 5 structural shell and outer Faraday cage.

SECTION VI

Internal Configuration

6.1 Observer Station and Session Structure

Parameter	Specification
Position	Geometric center: (1.5, 1.5, 0.0) m, seated head at $\sim$ (1.5, 1.5, 1.2) m
Seating	Non-magnetic meditation platform (wood/carbon fiber), no metal fasteners
EEG system	64-channel cap with optically-isolated amplifier, battery-only, fiber-optic data egress; all onboard radios (Bluetooth/Wi-Fi) hardware-disabled; emission verified < -80 dBm at airlock RF check
Physiological	Respiration band, ECG (heart rate and variability), surface EMG (muscle tone and micro-movement), and body-directed infrared/thermal channel (metabolic heat and skin temperature). All non-magnetic; recorded synchronously and pre-registered as nuisance covariates.
Comfort	Temperature-controlled seat pad (integrated with Layer 1 active thermal), adjustable headrest, emergency call button (fiber-optic)

Table 6.1: Observer station configuration.

Session timing is harmonized into a single structure used by both Phase 0 and Phase 1. Every condition has the same duration in both phases, eliminating fatigue-related variability across condition lengths. Phase 0 runs one condition per session; Phase 1 chains all seven conditions in a single session.

Block	Duration	Notes
Pre-condition buffer	5 min	Settling time after entry or after the previous condition transition. Data discarded.
C1, Active cognition	15 min	Standard condition window.
C2, Ordinary resting	15 min	Standard condition window.
C3, Focused attention	15 min	Standard condition window.
C4, Open monitoring	15 min	Standard condition window.
C5, Non-directed openness	15 min	Standard condition window.
C6, MPE (threshold)	15 min	Standard condition window. Gating state for C7 entry.
C7, Cessation	30 min	Extended window. Includes the four-step exit protocol for cessation.

Table 6.2: Standard session block structure (Phase 0 and Phase 1).

Phase 0, single-condition sessions. Each Phase 0 session consists of one consciousness condition only. Structure: empty-chamber calibration (10 min, before observer ingress), then observer entry plus 15-min ingress settling (per Section 3.2 step 6), then a 5-min pre-condition buffer, then the condition period (15 min for C1 to C6, 30 min for C7), then a brief post-session baseline. One session per observer per 24 h. Empty-chamber baseline measurements are run as separate sessions on the same randomized schedule.

Phase 0 total observer occupancy: ~35 min for C1 to C6 sessions (ingress settling + buffer + 15-min condition); ~50 min for C7 sessions (ingress settling + buffer + 30-min condition). Post-session baseline and exit add ~5 to 10 min.

Phase 1, chained multi-condition sessions. A single Phase 1 session may run all seven consciousness conditions in randomized order. Structure: empty-chamber calibration (10 min, before ingress), then observer entry plus 15-min ingress settling, then a 5-min pre-condition buffer plus condition period repeated for all seven conditions, then a brief post-session baseline. The empty-chamber baseline is measured in dedicated separate sessions, as in Phase 0.

Phase 1 total condition plus buffer duration: $6 \times (5 + 15) \text{ min} + 1 \times (5 + 30) \text{ min} = 155 \text{ min}$, plus the 15-min ingress settling and 10-min pre-ingress calibration = ~3 h total session length. C7 is positioned last in the condition order whenever the randomization permits, to avoid placing the exit protocol mid-chain. The condition-length structure (15 min for C1 to C6, 30 min for C7) is identical across phases: fatigue effects scale with total session occupancy, not with phase, and a consistent block structure simplifies cross-phase comparison and is consistent with the principle that the cognitive condition itself, not the time allocated to it, is the experimental variable.

6.2 Quantum Sensor Array

Parameter	Specification
Primary sensor (Phase 0)	Nitrogen-vacancy (NV) center in diamond, room-temperature T2 measurement
Primary sensor (Phase 1)	Superconducting transmon qubit, dilution refrigerator at 15 mK
Mounting	Non-magnetic optical table at center, 1.0 m above floor level
Distance to observer	1.0 to 1.5 m (adjustable along X-axis)
Secondary sensors	4x commercial QRNGs (ComScire PQ32MU), USB-to-fiber converters located externally where possible
Exploratory	Single-photon detectors (200 to 1000 nm), < 100 dark counts/s

Table 6.3: Quantum sensor array.

6.3 Environmental Monitoring Array

Instrument	Sensitivity	Rate	Purpose
3-axis fluxgate magnetometer	< 10 nT resolution	1 kHz	Residual magnetic field monitoring
SQUID magnetometer (LTS)	< 10 fT/sqrtHz @ 4 K (LHe)	10 kHz	Ultra-sensitive magnetic monitoring
SQUID magnetometer (HTS, alt.)	~50 fT/sqrtHz @ 77 K (LN2)	10 kHz	Lower-cost alternative if LHe infra unavailable
RF power meter	-80 dBm sensitivity	Continuous	Residual RF leakage detection
3-axis accelerometer	10 ⁻² g resolution	1 kHz	Vibration at sensor platform
Pt100 RTD sensors (x4)	+/- 0.001 K resolution	10 Hz	Temperature at 4 quadrants
Scintillator coincidence muon veto	Paired-panel coincidence; flags muon-coincident T2 events for exclusion	Continuous	Cosmic-ray muon rejection (Phase 0 standard channel)
Humidity sensor	+/- 0.5% RH	1 Hz	Interior humidity stability
Barometric sensor	+/- 0.1 hPa	1 Hz	Atmospheric pressure drift

Table 6.4: Environmental monitoring array.

All environmental channels are recorded synchronously with quantum sensor data and EEG via a common time base (GPS-disciplined 10 MHz reference, < 1 us inter-channel synchronization). Any anomaly in quantum data triggers automatic cross-correlation with all environmental channels.

SECTION VII

Support Systems

7.1 Power

- No AC mains inside chamber.
- Battery bank (48 V LiFePO₄, 5 kWh) in outer equipment room.
- DC-DC converters with optical isolation for each subsystem.
- All power lines filtered through EMI/RFI feedthrough filters at Layer 3 boundary.
- UPS backup for data acquisition (minimum 2 hours runtime).

7.2 Air Handling and Thermal Control

The air handling subsystem must balance three competing constraints: CO₂ removal for the full Phase 1 occupancy (up to ~3 h), < 0.1 m/s flow to preserve the acoustic environment, and removal of observer-generated heat.

Parameter	Specification / Rationale
CO ₂ load	Single adult at rest produces ~20 L/hr CO ₂ . For 3-hour Phase 1 session: ~60 L total.
Scrubber sizing	Soda-lime (Sofnolime 797) or molecular sieve, capacity at least 90 L CO ₂ for Phase 1, replaced every 2 sessions.
Air recirculation	Closed-loop HEPA-filtered, < 0.1 m/s at observer position, ducting through vibration-isolated bellows with acoustic baffles.
Fan	Brushless DC, vibration-isolated, located outside Layer 3. Fan vibration spectrum measured and included in performance budget (Appendix A).
Thermal control	Separate loop via Layer 1 active liner (see Section 5.1). Chilled glycol circulates through perforated inner liner; observer heat load absorbed at generation point.
O ₂ monitoring	Automatic abort if O ₂ < 19.5%. Dual-redundant sensors, one inside chamber, one in airlock.

Table 7.1: Air handling and thermal control.

7.3 Data Acquisition and Time Base

- All signals routed via single-mode fiber-optic feedthroughs (FC/APC connectors).
- Fiber penetrations through waveguide-below-cutoff tubes in Layers 2a, 2b, 3, and 5.
- External DAQ system: National Instruments PXIe chassis with multi-channel digitizers.
- GPS-disciplined 10 MHz oscillator. GPS antenna outside chamber, on roof. Reference distributed to interior via fiber through waveguide penetration. Never attempt GPS lock inside the shielded volume.

- Data storage: Redundant NVMe RAID array, minimum 10 TB.

7.4 Safety

- Emergency egress: Inner door operable from inside without tools (panic bar). Panic-bar events invalidate the session (Section 3.3).
- Redundant O2 monitoring with audible/visual alarm inside and outside.
- External video monitoring via fiber-optic camera (infrared, no EM emission inside chamber).
- Two-way fiber-optic intercom.
- Automatic session abort on any safety threshold breach.
- Fire suppression: Inert gas (Ar/N2) system, manual trigger only.

SECTION VIII

Quantum Sensor Integration

8.1 NV-Center in Diamond (Phase 0)

Parameter	Specification
Sensor type	Single NV center or NV ensemble in CVD diamond
T2 baseline	1 to 10 ms at room temperature (spin echo)
Optical excitation	532 nm laser, fiber-coupled, < 10 mW at diamond
Detection	Confocal fluorescence collection, 637 to 800 nm, APD detector
Microwave control	Coplanar waveguide on diamond mount, ~2.87 GHz
Readout rate	~10 ³ T2 measurements per session
Mounting	Diamond on PCB mount, on optical table, thermally stabilized
Advantages	Room temperature, no cryogenics, moderate cost, well-characterized
Limitations	Lower sensitivity to environmental perturbation than superconducting qubit

Table 8.1: NV-center in diamond (Phase 0 sensor).

NV sensor scope. The line item 'NV-center diamond sensor kit' in Section 10.3 covers the sensor module only (diamond, mount, optics, microwave source) at ~\$17,500 USD. The full sensor-platform figure (~\$30K to \$80K) additionally covers the optical table, integration mounting, calibration overhead, and platform-level vibration decoupling, all of which are budgeted under Structural (Section 10.2) and Labor (Section 10.5) line items in this document.

8.2 Superconducting Transmon Qubit (Phase 1 Upgrade)

Parameter	Specification
Sensor type	Transmon qubit in 3D aluminum cavity
T2 baseline	50 to 200 us at 15 mK
Operating temp	15 mK (dilution refrigerator required)
Readout	Dispersive readout via microwave cavity, HEMT amplifier chain
Cryostat	Dry dilution refrigerator (BlueFors LD or equiv.)
Footprint	1.0 m x 1.0 m floor, 2.0 m height (including vibration frame)
Cryogenic lines	Through dedicated vibration-isolated feedthroughs
Cryostat cost	\$234,000 to \$585,000 USD for cryostat alone; the full transmon-platform range (\$400K to \$800K) includes integration, shielding, and operator labor
Advantages	Maximum sensitivity to environmental perturbation
Limitations	Cost, complexity, specialized operators

Table 8.2: Superconducting transmon qubit (Phase 1 sensor).

8.3 QRNG Array

Parameter	Specification
Units	4x ComScire PQ32MU (32 Mbps each)
Interface	USB via fiber-optic USB extender; converter unit located outside chamber where possible. Any in-chamber converter shielded in local Faraday enclosure and included in Appendix A performance budget.
Mounting	Non-magnetic rack adjacent to optical table
Data volume	~14.4 GB per unit per 60-min session
Analysis	Entropy, autocorrelation, mean deviation; NIST SP 800-22 suite

Table 8.3: QRNG array.

SECTION IX

Mechanical Drawings (Descriptive)

Formal CAD drawings to be produced in SolidWorks or Fusion 360 during the detailed design phase. The following text descriptions define the required drawing set.

9.1 Drawing List

No.	Title	View	Content
TSP-001	Chamber Assembly, Plan View	Top-down, Z = 1.5 m section	All 5 layers (incl. dual mu-metal), airlock, feedthroughs, observer + sensor positions
TSP-002	Chamber Assembly, Elevation	South-facing section, Y = 1.5 m	Layer stack-up, floor structure, ceiling, height dimensions
TSP-003	Chamber Assembly, Cross Section	West-facing section, X = 1.5 m	Entry axis view, airlock detail, cable routing
TSP-004	Airlock Assembly	Plan + section detail	Dual-door arrangement, interlock, RF check station, seal details
TSP-005	Layer 2a/2b, Dual Mu-Metal	Exploded isometric	Panel layout, 50 mm air-gap standoffs, seam overlaps, annealing callouts
TSP-006	Layer 3, Copper Shell	Exploded isometric	Panel layout, weld seam map, grounding, RF gasket details
TSP-007	Layer 4, Isolation Platform	Plan + elevation	Isolator positions, frame, flexible coupling routes
TSP-008	Internal Layout	Isometric cutaway	Observer station, optical table, sensor mounts, cable routing, Layer 1 thermal manifold
TSP-009	Door Assemblies (Inner + Outer)	Section detail	Multi-layer seal stack, hinge, latch, panic bar, interlock wiring
TSP-010	Feedthrough Panel (West Face)	Section detail	Waveguide-below-cutoff tubes, fiber connectors, filter mounts, thermal fluid penetration
TSP-011	Foundation Plan	Plan view	Concrete slab, isolation trench, grounding grid, utilities

Table 9.1: Required drawing set.

9.2 Plan View Description (TSP-001)

Top-down cross-section at Z = 1.5 m (mid-height). Shows concentric square layers from outside in: Layer 5 aluminum shell (outermost, dark gray), Layer 4 isolation frame (hatched), Layer 3 copper shell (copper-colored), Layer 2b outer mu-metal (light gray), 50 mm air gap, Layer 2a inner mu-metal (light gray), Layer 1 acoustic/thermal panels (white) with thermal manifold overlay. Airlock extension on east face shown with dual doors and interlock annotation. Interior shows observer platform at center (circle, diameter 0.8 m), optical table 1.0 m west of center (rectangle, 1.5 m x 1.0 m), QRNG rack north of optical table. Feedthrough panel on west face. Observer entry path shown as dashed arrow from east through airlock to center. All dimensions in millimeters.

9.3 Elevation Description (TSP-002)

South-facing cross-section at $Y = 1.5$ m. Shows floor assembly (foundation slab, isolation mounts, floor plate), five functional layers (with 2a/2b shown as two distinct shells separated by air gap) rising vertically, ceiling assembly. Airlock visible on east side with dual doors. Observer seated figure (outline) at center facing west. Optical table with NV-center sensor at 1.0 m height, positioned west of observer. Overhead cable/fiber routing tray at 2.7 m. Air handling duct entry at ceiling level with acoustic baffle detail. Thermal fluid penetration shown on west face. Cryostat position indicated (Phase 1 dashed outline) with vibration isolation frame. All dimensions in millimeters.

SECTION X

Bill of Materials and Cost Estimates

All costs in USD for the Full Phase 0 configuration (NV-center sensor, LTS SQUID, no dilution cryostat). Costs assume single-unit procurement; volume discounts may apply for some items. Supplier quotes required for final budget.

10.1 Shielding Materials

Item	Qty	Unit Cost	Total	Notes
Mu-metal sheet, 3 mm (Layer 2a + 2b)	120 m ²	\$300/m ²	\$35,100	Dual-layer, incl. hydrogen annealing
OFHC copper sheet, 2 mm	60 m ²	\$150/m ²	\$8,400	C10100 grade
Aluminum plate, 5 mm	110 m ²	\$50/m ²	\$5,800	6061-T6
Copper mesh, 100/inch	110 m ²	\$20/m ²	\$2,300	Faraday layer
Acoustic foam panels	55 m ²	\$100/m ²	\$5,150	Basotect or equiv.
Aerogel blanket, 50 mm	55 m ²	\$200/m ²	\$9,650	Cryogel Z or equiv.
Perforated Al liner + thermal manifold	55 m ²	\$200/m ²	\$11,600	Integrated Peltier + fluid channels
RF finger stock gasket	20 m	\$50/m	\$1,400	BeCu, door + panels
Mu-metal finger stock (dual)	20 m	\$100/m	\$2,100	Both door shells
Air-gap standoffs (non-magnetic)	Lot		\$1,750	Between 2a and 2b
Subtotal, Shielding Materials			\$83,300	

Table 10.1: Shielding materials.

10.2 Structural and Mechanical

Item	Qty	Unit Cost	Total	Notes
Aluminum T-slot extrusion	250 m	\$20/m	\$4,400	80x80 mm (incl. airlock)
Pneumatic isolators	6 units	\$2,900 ea	\$17,550	TMC or Newport, uprated
Neoprene pads	12 units	\$200 ea	\$2,100	Floor isolation
Steel isolation frame	1 set	\$4,700	\$4,700	Welded box section
Optical table (granite)	1 unit	\$9,350	\$9,350	1.5 m x 1.0 m x 0.3 m
Outer door assembly (east)	1 unit	\$5,850	\$5,850	Multi-layer, outward-opening
Inner door assembly (east)	1 unit	\$7,600	\$7,600	Multi-layer with dual mu-metal
Door interlock system	1 unit	\$2,900	\$2,900	Electrical + mechanical
Airlock shielding extension	1 set	\$6,450	\$6,450	All layers continuous
Airlock RF check station	1 unit	\$1,750	\$1,750	RF meter + fluxgate
Feedthrough panels	2 units	\$2,550 ea	\$5,150	West face, waveguide tubes
Recirculating chiller + glycol loop	1 system	\$5,250	\$5,250	External, +/- 0.05 K output
Hardware / fasteners	Lot		\$2,350	Non-magnetic where required
Subtotal, Structural			\$75,400	

Table 10.2: Structural and mechanical.

10.3 Sensors and Instrumentation

Item	Qty	Unit Cost	Total	Notes
NV-center diamond sensor kit	1 unit	\$17,550	\$17,550	Sensor module only, see note in Section 8.1
ComScire PQ32MU QRNG	4 units	\$1,600 ea	\$6,300	32 Mbps each
64-ch EEG, optically-isolated	1 system	\$21,050	\$21,050	Radios disabled, fiber egress
SQUID magnetometer (LTS, 4 K)	1 unit	\$32,750	\$32,750	Single-channel LTS + LHe dewar
LHe dewar + transfer line	1 set	\$5,250	\$5,250	~30 L, weekly refill logistics
Fluxgate magnetometer	1 unit	\$3,500	\$3,500	3-axis
Single-photon detectors	2 units	\$7,000 ea	\$14,050	APD, 200 to 1000 nm
Scintillator coincidence muon veto	1 set	\$5,800	\$5,800	Paired-panel coincidence; standard Phase 0 channel
Pt100 RTD sensors	4 units	\$250 ea	\$950	0.001 K resolution
Accelerometer (3-axis)	1 unit	\$2,900	\$2,900	Seismic-grade
RF power meter	1 unit	\$1,750	\$1,750	Broadband
Subtotal, Sensors			\$111,850	

Table 10.3: Sensors and instrumentation.

10.4 Support Systems

Item	Qty	Unit Cost	Total	Notes
DAQ system (NI PXIe)	1 chassis + 4 cards	\$29,250	\$29,250	Multi-channel digitizer
GPS 10 MHz reference + fiber distrib	1 unit	\$2,900	\$2,900	Time base, external antenna
Battery bank (48 V, 5 kWh)	1 unit	\$3,500	\$3,500	LiFePO4
DC-DC converters + filters	Lot		\$2,350	Optically isolated
HEPA air handler + CO2 scrubber	1 system	\$6,450	\$6,450	Closed-loop, sized for Phase 1 occupancy
Fiber-optic USB extenders	6 units	\$450 ea	\$2,800	For QRNGs + EEG
Fiber cables + connectors	Lot		\$1,750	Single-mode, FC/APC
Safety systems (O2, intercom, camera)	Lot		\$3,500	Redundant
Workstation + storage	1 sys	\$4,700	\$4,700	Control room
Subtotal, Support			\$57,200	

Table 10.4: Support systems.

10.5 Labor and Integration

Item	Duration	Rate	Total	Notes
Chamber fabrication	35 days	\$600/day	\$20,500	Skilled metalworker + assistant
Shielding installation (dual)	20 days	\$700/day	\$14,050	Specialist RF/magnetic
Sensor integration	10 days	\$950/day	\$9,350	Quantum sensor specialist
SQUID + cryogenics integration	5 days	\$1,050/day	\$5,250	LHe system and vibration isolation
DAQ integration + calibration	10 days	\$700/day	\$7,000	Systems engineer
Commissioning + verification	12 days	\$700/day	\$8,400	Full-system testing
Subtotal, Labor			\$64,600	

Table 10.5: Labor and integration.

10.6 Budget Summary

Category	Cost (USD)
Shielding Materials	\$83,300
Structural and Mechanical (incl. airlock)	\$75,400
Sensors and Instrumentation	\$111,850
Support Systems	\$57,200
Labor and Integration	\$64,600
Contingency (15%)	\$58,850
TOTAL FULL PHASE 0 ESTIMATE	\$451,200

Table 10.6: Budget summary.

Scope and scaling. The Full Phase 0 estimate reflects the corrected engineering scope (dual mu-metal, active thermal, LTS SQUID, optically-isolated EEG, scintillator muon veto) and sits within the Phase 0 range of \$420K to \$470K, with the upper bound covering supplier-quote variance and final integration overhead. Three distinct configurations are recognized: (i) Micro-Pilot (\$2K to \$5K, QRNG desktop only, no chamber); (ii) Minimal Phase 0 (\$140K to \$210K, 1.5 m cube, single mu-metal, NV + QRNG only, no SQUID); (iii) Full Phase 0 (~\$451,200, as specified above).

SECTION XI

Fabrication Sequence

Total estimated fabrication timeline: 18 to 24 weeks from material procurement to first calibration session.

Phase	Duration	Activities	Dependencies
1. Procurement	8 to 12 weeks	Order mu-metal (both shells + annealing), copper, aluminum, isolators, sensors, SQUID, chiller	Budget approval
2. Foundation	1 to 2 weeks	Pour/verify reinforced concrete slab, install grounding grid	Site access
3. Outer shell	2 weeks	Assemble Layer 5 aluminum frame and panels, install Faraday mesh	Foundation complete
4. Vibration system	1 week	Install isolation frame, pneumatic isolators, passive dampers	Outer shell complete
5. Copper shell	1 to 2 weeks	Install Layer 3 copper panels, weld seams, install feedthroughs	Isolation frame stable
6. Outer mu-metal (2b)	1 week	Install Layer 2b panels (annealed), overlap seams, penetrations	Copper shell complete
7. Air-gap standoffs	0.5 week	Install non-magnetic standoffs establishing 50 mm radial gap	Layer 2b complete
8. Inner mu-metal (2a)	1 week	Install Layer 2a panels (annealed), overlap seams, coaxial penetrations	Standoffs complete
9. Acoustic + active thermal	1.5 weeks	Install aerogel, acoustic foam, Peltier liner, fluid manifold, perforated face	Layer 2a complete
10. Door + airlock	2 weeks	Install airlock structure, dual doors with dual mu-metal seals, interlock wiring	All layers complete
11. Internal fit-out	1 to 2 weeks	Optical table, sensor mounts, observer platform, cable routing	Chamber sealed

Phase	Duration	Activities	Dependencies
12. Systems integration	2 weeks	DAQ, power, air handling, SQUID cryogenics, chiller, safety, fiber	Internal fit-out
13. Commissioning	2 weeks	EM survey, magnetic field mapping, vibration spectra, thermal verification, T2 baseline	Full integration

Table 11.1: Fabrication sequence.

SECTION XII

Integration and Commissioning

12.1 Shielding Verification

- Electromagnetic: Sweep 10 Hz to 10 GHz with calibrated signal generator outside, spectrum analyzer inside. Confirm > 120 dB attenuation at all frequencies.
- Magnetic (DC): DC field mapping with fluxgate at 27 grid points (3x3x3). Confirm residual field < 50 nT.
- Magnetic (AC): AC magnetic sweep 0.1 to 100 Hz with explicit 50/60 Hz characterization. Confirm > 80 dB attenuation.
- Acoustic: Pink noise source outside at 90 dB SPL. Measure interior with calibrated microphone. Confirm < 20 dB SPL interior.
- Vibration: Measure floor vibration spectra with and without active isolation engaged. Confirm < 10^{-6} m/s² at sensor platform across 1 to 100 Hz.
- Thermal: 24-hour thermal recording with no occupant, then session-duration recording with occupant simulator (100 W thermal dummy). Confirm +/- 0.01 K stability at sensor position. Simulator session length should match longest planned occupancy (~3 h for Phase 1).

12.2 Quantum Sensor Baseline

- NV-center: Measure T2 (spin echo) with empty chamber. Establish baseline over minimum 100 measurements.
- QRNG: Run all 4 units for 24 hours with empty chamber. Run NIST SP 800-22 test suite.
- SQUID: 24-hour noise floor characterization, confirm sensitivity meets specification.
- Single-photon detectors: 24-hour dark count measurement. Confirm < 100 counts/s.

12.3 System Integration Test

- Simultaneous recording of all channels for 60 minutes with empty chamber.
- Verify time synchronization: < 1 us inter-channel offset.
- Verify data integrity: no dropped samples, no buffer overflows.
- Verify artifact detection pipeline: inject known environmental perturbation and confirm cross-correlation detection.

12.4 Ongoing Verification Schedule

Event	Actions Required
Annual	Full re-commissioning sweep (Section 12.1). Mu-metal degrades with mechanical stress and aging; annual magnetic remapping mandatory.
Post-relocation	Any physical move of the chamber invalidates prior commissioning. Full re-verification required.
Post-mechanical-shock	Any event involving chamber impact, drop, or seismic activity > MMI IV. Full re-verification.
Post-panic-bar event	Shielding continuity check minimum (EM sweep + magnetic sweep). Full re-verification if any anomaly found.
Session-to-session	Airlock RF check per Section 3.2 step 3. No recommissioning required if check passes.

Table 12.1: Ongoing verification schedule.

SECTION XIII

Phase Scaling Notes

Parameter	Micro-Pilot	Minimal P0	Full Phase 0	Phase 1
Inner volume	Desktop	1.5 m cube	3.0 m cube	4.0 to 5.0 m cube
Location	Any room	Quiet lab	University lab	Underground, at least 1,000 m depth
Shielding	None / portable RF	Single mu-metal + Cu	Dual mu-metal + Cu + thermal	5-layer + rock
Quantum sensor	None (QRNG only)	NV-center	NV-center	Superconducting transmon
QRNG	1x PQ4000KU	2x PQ32MU	4x PQ32MU	4x PQ32MU
EEG	Consumer (opt.)	32-ch wireless	64-ch optically-isolated	256-ch, fully optical
Env. monitors	Temp only	Basic (4 ch)	Full array + LTS SQUID	Full array + multi-ch SQUID
Budget (USD)	\$2K to \$5K	\$140K to \$210K	~\$451,200 (\$420K to \$470K range)	\$1M to \$3M
Timeline	2 to 4 weeks	3 to 4 months	6 to 9 months	2 to 3 years
Goal	Execution demo	Detect $d > 0.5$	Detect $d > 0.3$	Definitive test

Table 13.1: Phase scaling configurations.

Phase 1 budget. The Phase 1 range \$1M to \$3M USD covers the cryostat alone (\$234,000 to \$585,000, see Section 8.2), underground site prep, the full transmon platform (\$400K to \$800K), a 256-channel optically-isolated EEG, a multi-channel SQUID array, and 2 to 3 years of integration labor at SURF, SNOLAB, LNGS, or Boulby.

APPENDIX

Performance Budget

Expected contribution of each noise channel to T2 limitation and the isolation provided by the chamber. Design target: chamber-limited T2 exceeds intrinsic T2 by at least 10x.

Noise Channel	Unshielded	Attenuation	Residual	T2 Impact
RF / Microwave (10 kHz to 10 GHz)	~-30 dBm ambient	> 120 dB	< -150 dBm	Negligible
DC magnetic field	~50 uT (Earth)	> 80 dB (dual mu-metal)	< 5 nT	Below NV T2 limit
AC magnetic (50/60 Hz)	~100 nT lab	> 80 dB (dual mu-metal)	< 10 pT	Below NV T2 limit
Acoustic	~40 dB SPL	> 80 dB combined	< -40 dB SPL	Negligible
Vibration (1 to 100 Hz)	~10 ⁻³ m/s ²	> 60 dB (active+passive)	< 10 ⁻⁶ m/s ²	Negligible
Fan vibration (contribution)	Varies	Decoupled via bellows	< 10 ⁻⁶ m/s ²	Negligible
Thermal drift	+/- 1 K lab	Active control	+/- 0.01 K	Below NV T2 limit
In-chamber electronics	Variable	Local Faraday + airlock RF gate	< -80 dBm	Verified per session
Cosmic rays	~1/cm ² /min	None (Phase 0)	~1/cm ² /min	Flagged and vetoed in data

Table A.1: Performance budget by noise channel.

Cosmic ray note. Phase 0 (surface) does not attenuate cosmic-ray muons, which carry diurnal and barometric variation. Muon arrivals are detected by a scintillator coincidence veto (paired-panel coincidence, ~\$5,800) recorded as a standard channel, and muon-coincident T2 events are flagged and excluded from analysis. Phase 1 underground siting provides > 10² cosmic-ray attenuation via rock overburden, removing the background.

END OF DOCUMENT

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